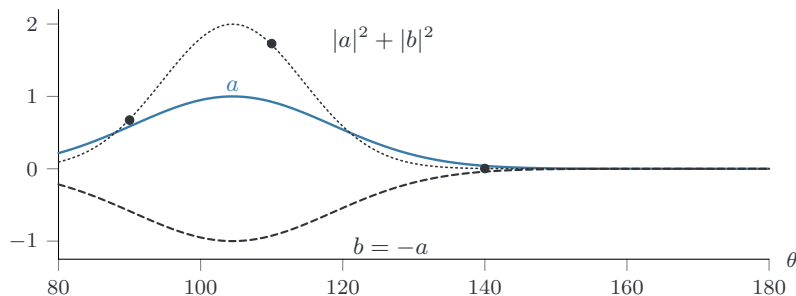
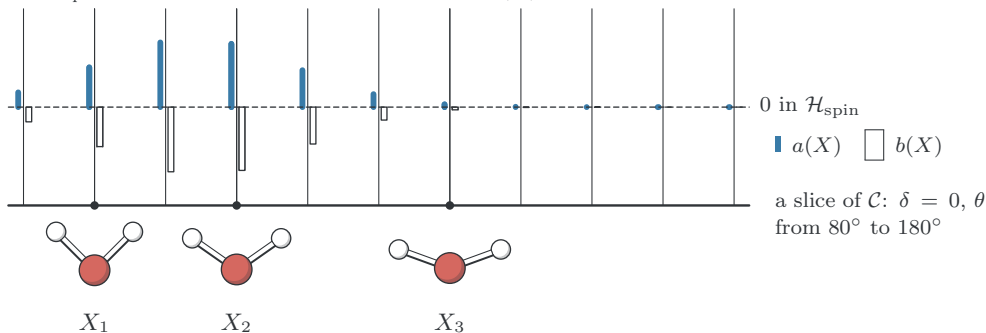


the same  $\mathcal{H}_{\text{spin}}$  over every  $X$ : on each fibre the value  $\Psi(X)$  as its two components  $a, b$



$$\Psi(X) = a(X)\xi + b(X)\eta$$

$$\|\Psi\|^2 = \int_{\mathcal{C}} (|a|^2 + |b|^2) dX$$

here  $b = -a$ :  
 $\Psi = a(\xi - \eta)$  is the proton singlet, as exchange requires of a rotation-invariant state on this slice

$\xi = \uparrow\downarrow \omega, \eta = \downarrow\uparrow \omega$  orthonormal in  $\mathcal{H}_{\text{spin}} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^1$ ;  $a(\theta)$  is an illustrative profile  
the dots are  $\|\Psi(X_k)\|^2$